

Week 10 - Mock exam 2

1) Determine whether the following statements are true or false.

(a) The triple, $(\mathbb{Z}, \times, 1)$, given by set of integers, $\mathbb{Z}$, multiplication, and $1 \in \mathbb{Z}$, defines a group.

(b) $10,000,000,004 \equiv 5 \pmod{11}$.

(c) The polynomial $123x^{456} + 789x - 123456789$ has at least one complex root.

(d) If the linear system,

$$\begin{aligned} ax + by &= t_0 \\ cx + dy &= t_1, \end{aligned}$$

has more than one solution, then the linear system,

$$\begin{aligned} ax + by &= s_0 \\ cx + dy &= s_1, \end{aligned}$$

also has more than one solution.

(e) The function $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = 2x$ is a group homomorphism from $(\mathbb{Z}, +, 0)$ to itself.

(f) The polynomial $123x^{456} + 789x^{123456789} - 912$ has at least one real root.

(g) Let $\mathbb{R}_1^\times = (\mathbb{R}_{>0}, \times, 1)$ be the group of positive real numbers under multiplication. The function from $\mathbb{R}_1^\times$ to itself defined by $f(x) = 2x$ is a group isomorphism.

(h) Let $\mathbb{R} = (\mathbb{R}, +, 0)$ be the group of real numbers under addition. The function from $\mathbb{R}$ to itself defined by $f(x) = 2x$ is a group isomorphism.

(i) If the linear system,

$$\begin{aligned} ax + by &= t_0 \\ cx + dy &= t_1, \end{aligned}$$

has a unique solution, then the linear system,

$$\begin{aligned} ax + by &= s_0 \\ cx + dy &= s_1, \end{aligned}$$

also has a unique solution.

(j) If the linear system,

$$\begin{aligned} ax + by &= t_0 \\ cx + dy &= t_1, \end{aligned}$$

has a solution, then the linear system,

$$\begin{aligned} ax + by &= s_0 \\ cx + dy &= s_1, \end{aligned}$$

also has a solution.

2) Let G be a group, and let $g \in G$. Prove that the set $\{g^n : n \in \mathbb{Z}\}$ is a group with the same multiplication and unit, where we use the notational convention that any element of a group to the 0th power is the identity element, and for $n > 0$, $g^{-n} := (g^{-1})^n$.

3) For $n \in \mathbb{Z}_{\geq 1}$, let $\mathbb{Z}/n = \{0, \dots, n-1\}$ denote the group whose operation is given by addition modulo n (and whose identity element is 0).

(a) Prove that for any n, m , the function $\mathbf{0} : \mathbb{Z}/n \rightarrow \mathbb{Z}/m$ defined by $\mathbf{0}(k) = 0$ for all $k \in \mathbb{Z}/n$ is a group homomorphism.

(b) Prove that there exists group homomorphisms $f : \mathbb{Z}/3 \rightarrow \mathbb{Z}/12$ and $g : \mathbb{Z}/12 \rightarrow \mathbb{Z}/3$, such that neither f nor g are equal to $\mathbf{0}$ above.

(c) Prove that the only group homomorphism $\mathbb{Z}/2 \rightarrow \mathbb{Z}/3$ is $\mathbf{0}$.

4) Consider the linear system,

$$\begin{aligned}2x - 6y &= 5 \\4x - 12y &= 7.\end{aligned}$$

(a) Prove whether the system has zero, one, or more than one solution.

(b) Does there exist zero, one, or more than one pair $(t_0, t_1) \in \mathbb{R}^2$ where $(t_0, t_1) \neq (5, 7)$, and such that the linear system,

$$\begin{aligned}2x - 6y &= t_0 \\4x - 12y &= t_1,\end{aligned}$$

has a solution? Justify your answer. If your answer is not zero, describe (all) the pair(s), (t_0, t_1) , so that the system has a solution.

(c) Does there exist zero, one, or more than one pair $(t_0, t_1) \in \mathbb{R}^2$ where $(t_0, t_1) \neq (5, 7)$, and such that the linear system,

$$\begin{aligned}2x - 6y &= t_0 \\4x - 12y &= t_1,\end{aligned}$$

has a unique solution? Justify your answer. If your answer is not zero, describe (all) the pair(s), (t_0, t_1) , so that the system has a unique solution.